

**SECTION A**

General Unit Information:

Unit Name	Fundamentals Of Software Engineering
Unit Code	CIS092-1
Level	4
Credit Value	30
Location(s) of Delivery	On Campus
Period(s) of Delivery	15 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	This unit has two elements, the first is aimed at your professional development and the second introduces you to the practices of the software engineering and related emerging technologies. Firstly, this unit will equip you with the skills needed to cope with the demands of your course (which are also important to employers) and to help you to become more 'self-aware' learners. It will also teach you to develop an awareness of the expectations and professional requirements of your chosen industry. The second element of the unit will lead you to develop your fundamental skills of software engineering, and the ability to use them in practise. Furthermore, by exposing students to state-of-the-art applicable emerging technologies, this section will equip every student with a solid and forward vision within the profession of Software Engineering.

Syllabus Content	Fundamentals of Study and Employability • Introduction – making sense of academic life: support mechanisms, AUPs, codes of conduct, academic rules and regulations, expectations - why are you here? • How to Study – Learning Styles, Note Taking, use of VLE, mind mapping, time management • Career management skills, the notion of employability, skills and competences and knowledge relevant to particular industries and job roles, assessment centres, Career goal setting, Action planning, sources of careers information. Developmental opportunities: professional bodies, volunteering and job shop • Making sense of the skills and knowledge to be developed - a roadmap for your course – Bloom's taxonomy • Obtaining resources in the library and online, evaluating quality of sources. Referencing, paraphrasing and plagiarism. • Writing styles- formal report v reflective, first person, third person. Report structure • Group work; Belbin group roles and group dynamics, creativity techniques. group decision making and problem solving, effective behaviours • Oral presentations, effective behaviours • Commercial awareness – finding out about a particular company, finding out about the wider industry • Basic legislation relevant to Computing and Engineering industries, Data Protection Act, Disabilities Discrimination Act. • Ethics relating to Computing and Engineering industries. Codes of conduct, Health and Safety in the workplace. Fundamentals of Software Engineering Focused on but not limited to: <i>Overview and introduction of computer hardware, resource management and operating systems</i> • O/S history, functionality • Modern operating systems and applications • Memory, process, IO and file management • Practices with Linux, dockers and virtual machines <i>Emerging Technologies on Computer Science such as</i> • Emerging technologies trends • Google APIs for application developments • Open source software such as TensorFlow etc. for Artificial Intelligence • Git for distributed version controls
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Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Describe and identify the application of software engineering in a range of contexts applicable to the course of study

2. Demonstrate the following skills and abilities

Apply the practical and professional skills needed to cope with the demands of your course (which are also important to employers) and demonstrate an ability to reflect on your development

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	72	57	132	39	0	300
Percentage	24%	19%	44%	13%	0%	100%

Approach to Learning

This unit will be delivered using formal lectures, lab-based practical exercises, and tutorial sessions to help students to understand the fundamentals of software engineering and develop their problem-solving and professional skills. The following specific elements are included:

- Lectures introduce the practices of the software engineering in relation to business environments and their operations, and the professional processes of software engineering development
- Lab-based practical sessions and tutorials provide students with opportunities of consolidating the concepts and techniques introduced in the lectures with the aim of applying them to develop solutions to problems based on real-world scenarios.
- Guided and self-directed review of published literature, e.g. books, research papers, web-based resources including tutorials on the tools and techniques, and products used for software engineering applications. The use of electronic and paper based resources is expected and encouraged.

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Coursework - Individual Report	1 and 2	30	7	1000 words
2	PR-Oral	1 and 2	70	12	10 minutes (presentation)

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	You will research and report on skills requirements in a relevant industry
Contextual Understanding, including where employer based	You will identify and explore relevant bodies and events within your industry to help develop your commercial awareness

Collaboration	You will work as part of a group to develop and present a solution to given problems
Enterprise	You will be asked to analyse your own strengths and develop an action plan to help you achieve your objectives
Assessment 1 Details	You will submit an individual written report of 1000 words relating to core concepts of computer science and software development.
Assessment 2 Details	You will deliver a 10 minute presentation covering core concepts of computer science, software development and computer law.
Assessment 3 Details	
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

Identify at least three relevant sources of information for your report.

Use these sources to present a summary of the topic.

Write a justification for the choices you have made in relation to a chosen topic.

In order to pass Assessment 2 you will need to:

Explain the aspects of software engineering.

Explain different approaches, frameworks and architectures in computer science and software engineering.

Explain the processes of design and analysis of computing systems logically.

In order to pass Assessment 3 you will need to:

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

See <https://rl.talis.com/3/bedfordshire/lists/6BA75309-128E-9697-1CF3-EB20C419201A.html>

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

See <https://rl.talis.com/3/bedfordshire/lists/6BA75309-128E-9697-1CF3-EB20C419201A.html>

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

See <https://rl.talis.com/3/bedfordshire/lists/6BA75309-128E-9697-1CF3-EB20C419201A.html>

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	Y	
The approach to practical work takes account of the needs of students with disabilities?	Y	
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	
The assessment tasks provide all students with an equal opportunity to succeed?	Y	

Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	NA	
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* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100374
Unit Co-ordinator	Tess Crosbie
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Software Engineering